

Real-Time Campus Aeroallergen Dispersion Modeling Using Gaussian Plume Theory and Satellite-Based Vegetation Detection

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Abstract

We present Campus AeroAllergen Mapping (CAM), a web-based system for modeling hyperlocal pollen dispersion across a school campus at meter-scale resolution. The framework integrates a steady-state Gaussian plume model with Schulman-Scire building downwash corrections to compute pollen concentration at student breathing height ($z = 1.5$ m). Pollen sources are automatically identified from high-resolution satellite imagery using color-based segmentation with watershed splitting, refined through iterative human-in-the-loop calibration. Temporal emission variability is captured by species-specific Gaussian phenological gate functions. Building footprints detected from satellite imagery define indoor exclusion zones where concentration is zeroed. Applied to a 25-hectare high school campus containing 1,018 verified trees across 6 allergenic genera, the system produces real-time concentration fields at 5.3 m spatial resolution with sub-second computation time. A fixed annual color scale enables intuitive seasonal comparison, revealing that summer concentrations are approximately 6% of the spring peak. The system provides actionable path-integrated exposure assessments for allergy-sensitive students.

Keywords: Gaussian plume model, pollen dispersion, aeroallergen mapping, building wake effects, satellite vegetation detection, phenological modeling

1 Introduction

Allergic rhinitis affects 10–30% of the global population, with airborne pollen constituting the dominant environmental trigger [Bousquet et al., 2008]. On school campuses where students transit between buildings through corridors lined with allergenic flora, localized pollen concentrations can vary by orders of magnitude over distances of tens of meters due to source proximity, wind channeling, and structural aerodynamic effects.

Existing pollen monitoring relies on regional stations reporting area-averaged counts at temporal resolutions of hours to days [Sofiev et al., 2013]. These measurements cannot resolve the hyperlocal concentration gradients determining individual exposure during a 5-minute walking transit. We address this gap by developing a physics-based framework that:

1. Solves the atmospheric advection-diffusion equation analytically at meter-scale resolution;
2. Accounts for building-induced flow modifications using empirical downwash models;
3. Automatically identifies pollen sources from satellite imagery;

4. Modulates emissions temporally using species-specific phenological functions;
5. Masks indoor building volumes from the exposure field.

The system is deployed at Henry M. Gunn High School (Palo Alto, CA), a 25-hectare campus containing over 1,000 trees spanning multiple allergenic genera.

2 Atmospheric Dispersion Model

2.1 Physical Picture

When a tree releases pollen grains into the air, two physical processes determine where those grains end up:

1. **Advection** (bulk transport by wind): The wind carries the pollen cloud bodily in the downwind direction, like a river carrying a dye patch. If the wind blows from the southwest at 3 m/s, the pollen cloud moves northeast at 3 m/s.
2. **Diffusion** (turbulent spreading): Atmospheric turbulence—random gusts and eddies—causes the pollen cloud to spread out in all directions as it travels. The cloud starts concentrated near the tree and becomes progressively wider and more dilute downwind, forming a cone-shaped plume.

The combination produces a characteristic pattern: highest concentration immediately downwind of the tree, decreasing with distance as the plume spreads, and zero concentration upwind.

2.2 Governing Equation

These processes are described mathematically by the *advection-diffusion equation*:

$$\underbrace{\frac{\partial C}{\partial t}}_{\text{concentration change}} + \underbrace{u \frac{\partial C}{\partial x} + v \frac{\partial C}{\partial y}}_{\text{wind carries pollen (advection)}} = \underbrace{K_x \frac{\partial^2 C}{\partial x^2} + K_y \frac{\partial^2 C}{\partial y^2} + K_z \frac{\partial^2 C}{\partial z^2}}_{\text{turbulence spreads pollen (diffusion)}} - \underbrace{\lambda C}_{\text{settling/removal}} + \underbrace{Q \delta(\mathbf{x} - \mathbf{x}_s)}_{\text{tree emitting pollen}}$$

(1)

Each term has a clear physical meaning:

- $C(x, y, z, t)$ — **Pollen concentration** (grains/m³) at position (x, y, z) and time t . This is what a student breathes in.
- (u, v) — **Wind velocity components** (m/s). u is the east-west wind speed, v is the north-south wind speed. Measured from weather station or API. A 4 m/s southwest wind means $u \approx 2.8$ m/s (eastward) and $v \approx 2.8$ m/s (northward).
- K_x, K_y, K_z — **Turbulent eddy diffusivity** (m²/s). These quantify how vigorously the atmosphere mixes. On a calm, stable morning ($K \approx 1$ m²/s), pollen stays concentrated near the source. On a windy, sunny afternoon ($K \approx 50$ m²/s), turbulent eddies rapidly dilute the pollen cloud. K_z (vertical) is typically smaller than K_x, K_y (horizontal) because vertical mixing is suppressed by gravity.

- λ — **Deposition/removal rate** (s^{-1}). Pollen is continuously removed from the air by gravitational settling onto the ground, impaction on surfaces, and washout by rain. Heavier pollen (Pine, $60 \mu\text{m}$ diameter) has λ about $5\times$ larger than light pollen (Oak, $20 \mu\text{m}$), meaning Pine falls out of the air much faster.
- Q — **Source emission rate** (grains/s). How many pollen grains the tree releases per second. A mature Oak at peak bloom emits ~ 500 grains/s from its canopy.
- $\delta(\mathbf{x} - \mathbf{x}_s)$ — **Dirac delta function**. Indicates that emission occurs at a single point $\mathbf{x}_s$ (the tree’s location). This is the “point source” approximation.

2.3 Gaussian Plume Solution

Solving the full PDE numerically at every timestep would be computationally expensive. Instead, we use the *steady-state* analytical solution (assuming wind conditions change slowly relative to pollen transport time). This yields the famous **Gaussian plume** formula [Pasquill, 1961]:

$$C(x', y', z) = \underbrace{\frac{Q}{2\pi U \sigma_y \sigma_z}}_{\text{dilution factor}} \cdot \underbrace{\exp\left(-\frac{y'^2}{2\sigma_y^2}\right)}_{\text{crosswind spread}} \cdot \underbrace{V(z)}_{\text{vertical profile}} \quad (2)$$

Each factor has a physical interpretation:

- $\frac{Q}{2\pi U \sigma_y \sigma_z}$ — **Dilution factor**. The faster the wind (U large) and the wider the plume (σ_y, σ_z large), the more diluted the pollen becomes. Doubling wind speed halves the concentration everywhere.
- $\exp(-y'^2/2\sigma_y^2)$ — **Crosswind Gaussian profile**. Concentration is highest on the plume centerline ($y' = 0$, directly downwind) and drops off as a bell curve to either side. σ_y is the “width” of this bell curve—how far sideways the pollen has spread.
- $V(z)$ — **Vertical profile with ground reflection**:

$$V(z) = \underbrace{\exp\left(-\frac{(z-H)^2}{2\sigma_z^2}\right)}_{\text{direct path from source}} + \underbrace{\exp\left(-\frac{(z+H)^2}{2\sigma_z^2}\right)}_{\text{ground-reflected path}} \quad (3)$$

The first term is pollen arriving directly from the tree canopy at height H . The second term accounts for pollen that has hit the ground and “bounced” back up (mathematically modeled as a mirror image source below ground). This ensures no pollen is lost through the ground surface.

2.4 Key Parameters Explained

2.4.1 σ_y and σ_z : Plume Spread (meters)

These are the most important parameters. They control how wide (σ_y) and tall (σ_z) the plume is at any given downwind distance:

- **Near the tree** ($x' = 10 \text{ m}$): $\sigma_y \approx 0.8 \text{ m}$, $\sigma_z \approx 0.6 \text{ m}$. The plume is narrow—only a few meters wide. Concentration is very high.

- **50 m downwind:** $\sigma_y \approx 3.5$ m, $\sigma_z \approx 2.5$ m. The plume has spread to about 7 m wide.
- **200 m downwind:** $\sigma_y \approx 12$ m, $\sigma_z \approx 8$ m. The plume is now 24 m wide but much more dilute.

They grow with distance because turbulence has more time to mix the pollen sideways and vertically. The growth rate depends on *atmospheric stability*:

$$\sigma_y = a_y \cdot (x')^{b_y}, \quad \sigma_z = a_z \cdot (x')^{b_z} \quad (4)$$

2.4.2 Atmospheric Stability Classes (A–F)

Stability describes how much the atmosphere resists or encourages vertical mixing:

- **Class A (very unstable):** Hot sunny afternoon. Strong updrafts vigorously mix pollen vertically. Plume spreads rapidly, diluting quickly. *Paradoxically, unstable conditions produce lower ground-level concentrations because pollen is lofted upward.*
- **Class D (neutral):** Overcast, moderate wind. Default condition. Moderate mixing.
- **Class F (very stable):** Clear, calm night or early morning. Very little vertical mixing. Plume stays narrow and concentrated near emission height. *Worst case for ground-level exposure if source is low.*

We estimate stability from two observable quantities:

- Wind speed < 2 m/s + clear sky $\rightarrow$ Class A/B (unstable, strong solar heating)
- Wind speed 3–5 m/s + partly cloudy $\rightarrow$ Class C/D (neutral)
- Wind speed > 6 m/s $\rightarrow$ Class D (mechanical turbulence dominates)

Figure 1: Gaussian Plume Dispersion Under Varying Conditions
($Q = 1000$ grains/s, $H = 6$ m, $z = 1.5$ m, wind from West)

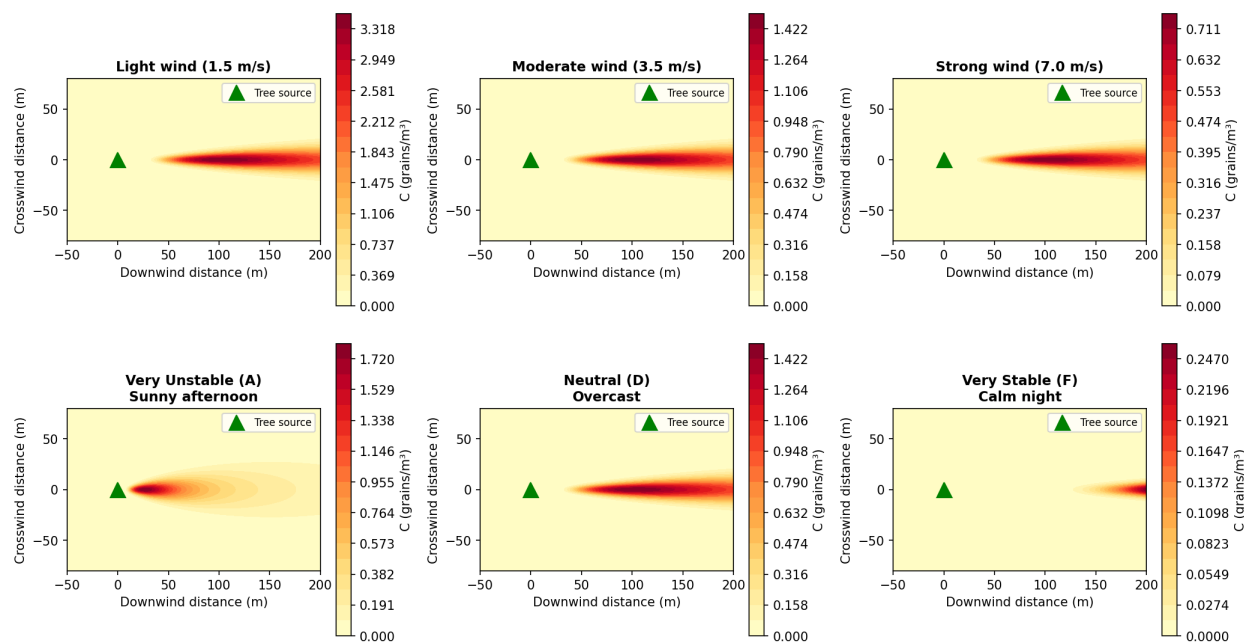


Figure 1: Gaussian plume concentration fields under varying conditions. Top row: effect of wind speed (stability class D). Stronger winds dilute and elongate the plume. Bottom row: effect of atmospheric stability (wind = 3.5 m/s). Unstable conditions (A) spread pollen rapidly; stable conditions (F) confine it in a narrow, concentrated plume.

2.4.3 U : Wind Speed (m/s)

Mean horizontal wind speed at reference height (typically 10 m). Obtained from OpenWeatherMap API. Physical effects:

- **Strong wind** ($U > 5$ m/s): Pollen is rapidly carried far downwind but heavily diluted. Concentration drops quickly with distance.
- **Light wind** ($U < 2$ m/s): Pollen lingers near the source tree. High concentrations in a small area. The model sets a minimum of $U = 0.5$ m/s to avoid division by zero (calm conditions are handled as very light drift).

2.4.4 H : Effective Source Height (meters)

The height at which pollen enters the atmosphere. For trees, this is the canopy height where catkins/flowers release grains. We use $H = 6.0$ m (average for campus trees ranging from 4 m shrubs to 15 m oaks). The concentration at ground level ($z = 1.5$ m) depends on how far pollen must fall/mix downward from height H .

2.4.5 z : Receptor Height (meters)

Fixed at $z = 1.5$ m—the height of a student's nose and mouth (breathing zone). This is where inhaled dose matters for allergic response.

2.4.6 (x', y') : Wind-Aligned Coordinates

The plume always points downwind. To compute concentration at any campus location, we transform from map coordinates to a frame aligned with the current wind direction:

$$x' = \Delta x \cos \alpha + \Delta y \sin \alpha \quad (\text{downwind distance from tree}) \quad (5)$$

$$y' = -\Delta x \sin \alpha + \Delta y \cos \alpha \quad (\text{crosswind offset}) \quad (6)$$

where $\alpha = \frac{\pi}{180}(270^\circ - \theta)$ converts the meteorological wind direction θ (degrees clockwise from North) to a mathematical angle. Key consequences:

- If you are directly downwind of a tree ($y' = 0$): maximum exposure.
- If you are perpendicular to the wind ($x' = 0$): zero concentration (upwind or crosswind).
- If the wind shifts direction, the entire plume rotates to point the new way.

2.5 Humidity Effects on Pollen Dispersion

Relative humidity (RH) affects pollen transport through multiple mechanisms that our model incorporates:

2.5.1 Hygroscopic Swelling

Pollen grains are hygroscopic—they absorb water vapor from humid air and swell. The effective aerodynamic diameter increases with humidity following a growth factor:

$$d_p(RH) = d_{p,dry} \cdot \left(1 + \kappa \cdot \frac{RH}{100 - RH} \right)^{1/3} \quad (7)$$

where κ is the species-specific hygroscopicity parameter ($\kappa \approx 0.05$ – 0.15 for pollen). At 80% RH, Oak pollen swells from 22 μm to approximately 25 μm , increasing its settling velocity by $\sim 30\%$:

$$v_s(RH) = v_{s,dry} \cdot \left(\frac{d_p(RH)}{d_{p,dry}} \right)^2 \cdot \frac{\rho_p(RH)}{\rho_{p,dry}} \quad (8)$$

The density ratio accounts for water uptake reducing the effective density. The net effect: humid conditions cause pollen to settle faster, reducing far-field concentrations but increasing deposition near the source.

2.5.2 Humidity-Dependent Emission Suppression

Most tree species suppress pollen release under high humidity. Anthers (the pollen-bearing structures) require drying to dehisce (split open). The emission rate is modulated by a humidity factor:

$$Q_i(t, RH) = Q_i(t) \cdot f_{RH}(RH) \quad (9)$$

where:

$$f_{RH}(RH) = \begin{cases} 1.0 & RH < 50\% \\ 1.0 - 0.8 \cdot \frac{RH - 50}{50} & 50\% \leq RH < 90\% \\ 0.1 & RH \geq 90\% \end{cases} \quad (10)$$

At 50% RH, emission is unrestricted. At 90% RH (foggy morning), emission drops to just 10% of the dry-air maximum. This explains why pollen counts are typically highest on dry, warm afternoons and lowest during humid mornings or after rainfall.

2.5.3 Osmotic Rupture and Sub-Pollen Particles

When relative humidity rapidly exceeds $\sim 95\%$ (e.g., during thunderstorms), pollen grains can undergo *osmotic rupture*: the grain absorbs water faster than its wall can expand, causing it to burst and release hundreds of sub-pollen particles (SPP) of $0.5\text{--}4\ \mu\text{m}$ diameter. These SPPs:

- Are small enough to penetrate deep into the lower respiratory tract (bronchi and alveoli), unlike intact grains which deposit in the nose/throat
- Have negligible settling velocity ($v_s < 0.01\ \text{cm/s}$), remaining airborne for hours
- Carry the same allergenic proteins as the parent grain
- Are implicated in “thunderstorm asthma” events

We model this as a conditional burst event. If $RH > 95\%$ and $\Delta RH/\Delta t > 10\%/hr$ (rapid humidity spike):

$$Q_{SPP,i}(t) = N_{SPP} \cdot Q_i(t) \cdot f_{rupture} \quad (11)$$

where $N_{SPP} \approx 700$ is the number of sub-pollen particles per ruptured grain and $f_{rupture} \approx 0.3$ is the fraction of airborne grains that rupture. The SPPs are then modeled as a separate Gaussian plume with near-zero settling velocity and greatly extended atmospheric lifetime.

2.5.4 Wet Deposition (Rainfall Washout)

Rain removes pollen from the atmosphere through below-cloud scavenging. The washout rate is modeled as an additional first-order removal term:

$$\lambda_{wet} = \Lambda \cdot P^{0.7} \quad (12)$$

where P is the rainfall intensity (mm/hr) and $\Lambda \approx 5 \times 10^{-4}\ \text{s}^{-1}/(\text{mm/hr})^{0.7}$ is the scavenging coefficient for pollen-sized particles. During moderate rain (5 mm/hr), washout removes pollen with an effective half-life of ~ 20 minutes, rapidly clearing the atmosphere. The total removal rate becomes:

$$\lambda_{total} = \lambda_{dry} + \lambda_{wet} \quad (13)$$

2.5.5 Implementation

Our system obtains current relative humidity from the weather API (OpenWeatherMap provides RH as a standard field). The humidity corrections are applied as:

1. Emission rate adjusted by f_{RH} (Eq. 9)
2. Settling velocity adjusted by hygroscopic swelling (Eq. 7)
3. SPP burst event triggered if RH exceeds 95% with rapid onset
4. Wet deposition added during active precipitation

On a typical Palo Alto afternoon ($RH \approx 40\text{--}55\%$), humidity corrections are minimal. During marine-layer mornings ($RH \approx 80\text{--}90\%$), emission is suppressed by 50–80%, significantly reducing campus pollen load—explaining why early-morning commuters often experience fewer symptoms than those arriving mid-morning after the fog burns off.

Figure 4: Humidity Effects on Pollen Behavior

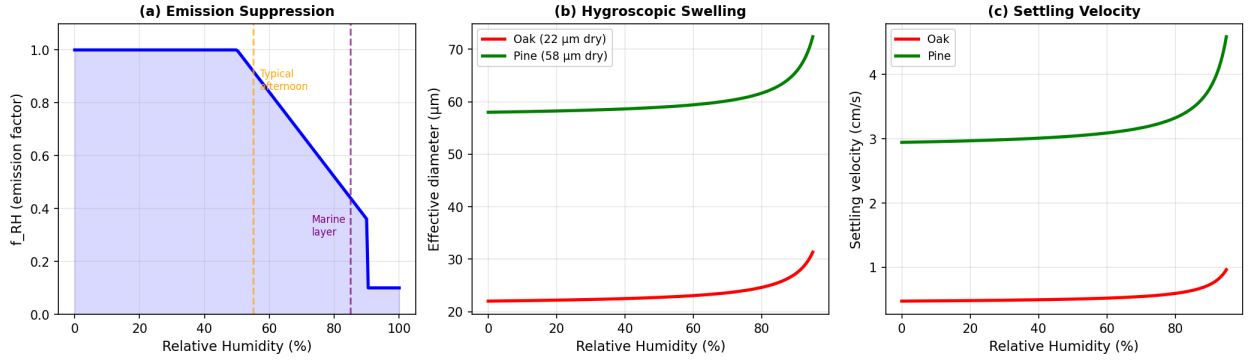


Figure 2: Humidity effects on pollen behavior. (a) Emission suppression factor f_{RH} : pollen release drops sharply above 50% RH. (b) Hygroscopic swelling increases grain diameter at high humidity. (c) Resulting increase in gravitational settling velocity.

2.6 Species-Dependent Particle Dynamics

Pollen grains vary significantly in aerodynamic properties across species, affecting their atmospheric transport behavior. Three key species-dependent parameters modify the dispersion model:

2.6.1 Gravitational Settling Velocity

Pollen grains are not neutrally buoyant; they settle under gravity at a terminal velocity v_s governed by Stokes' law for spherical particles in laminar flow:

$$v_s = \frac{\rho_p d_p^2 g}{18 \mu} \quad (14)$$

where ρ_p is the pollen grain density (kg m^{-3}), d_p is the aerodynamic diameter (m), $g = 9.81 \text{ m s}^{-2}$ is gravitational acceleration, and $\mu = 1.81 \times 10^{-5} \text{ Pa}\cdot\text{s}$ is the dynamic viscosity of air at 20°C .

Table 1 lists the physical properties and resulting settling velocities for each campus species. Pine pollen, with its large diameter ($\sim 60 \mu\text{m}$), settles at 3.1 cm/s —roughly $6\times$ faster than the small Oak grains ($\sim 22 \mu\text{m}$, 0.5 cm/s). This means Pine pollen deposits closer to the source tree, while Oak pollen remains airborne longer and disperses further downwind.

Table 1: Pollen grain physical properties and settling velocities.

Species	d_p (μm)	ρ_p (kg/m^3)	v_s (cm/s)	Airborne half-life
Valley Oak	22	1050	0.5	Long (hours)
Coast Live Oak	20	1050	0.4	Long (hours)
Pine	58	550	3.1	Short (minutes)
Redwood	30	800	1.0	Moderate
Chinese Elm	28	1000	1.1	Moderate
Sycamore	20	1020	0.4	Long (hours)

2.6.2 Tilted Plume: Gravitational Modification

The settling velocity tilts the plume centerline downward. The effective plume height decreases with downwind distance:

$$H_{eff}(x') = H - \frac{v_{s,i} \cdot x'}{U} \quad (15)$$

where H is the initial emission height, $v_{s,i}$ is the species-specific settling velocity, x' is the downwind distance, and U is the wind speed. For heavy Pine pollen in light winds ($U = 2$ m/s), the plume centerline drops by 1 m over just 65 m of travel, causing rapid ground-level concentration increase near the source.

Substituting into the Gaussian plume (Eq. 2), the species-dependent vertical term becomes:

$$V_i(z, x') = \exp\left(-\frac{(z - H_{eff,i})^2}{2\sigma_z^2}\right) + \exp\left(-\frac{(z + H_{eff,i})^2}{2\sigma_z^2}\right) \quad (16)$$

2.6.3 Deposition Velocity and First-Order Decay

Pollen removal from the atmosphere occurs through gravitational settling (dry deposition) and impaction on surfaces. The effective decay rate λ_i in Eq. (1) is species-dependent:

$$\lambda_i = \frac{v_{d,i}}{H_{mix}} \quad (17)$$

where $v_{d,i}$ is the total deposition velocity (combining gravitational settling and surface impaction) and H_{mix} is the mixing height. Larger grains have higher deposition velocities, resulting in faster atmospheric removal:

$$v_{d,i} = v_{s,i} + v_{imp,i} \quad (18)$$

The impaction velocity v_{imp} depends on grain inertia (Stokes number) and surface roughness. For campus-scale transport over grass and pavement, typical values are $v_{imp} \approx 0.2$ – 0.5 cm/s for small grains and 1 – 2 cm/s for large grains.

2.6.4 Turbulent Diffusivity

The lateral (K_y) and vertical (K_z) eddy diffusivities in Eq. (1) are properties of the atmosphere, not the particle. However, the *effective* vertical diffusivity for heavy particles is reduced because gravitational settling counteracts upward turbulent mixing:

$$K_{z,eff,i} = K_z \cdot \exp\left(-\frac{v_{s,i}}{u_*}\right) \quad (19)$$

where $u_* \approx 0.1$ – 0.4 m/s is the friction velocity. For small Oak pollen ($v_s = 0.5$ cm/s $\ll u_*$), the correction is negligible ($K_{z,eff} \approx K_z$), and the particle behaves as a passive tracer. For heavy Pine pollen ($v_s = 3.1$ cm/s), the effective diffusivity is reduced by $\sim 10\%$, confining the plume closer to the ground.

2.6.5 Species-Specific Plume Behavior Summary

Combining these effects, the full species-dependent Gaussian plume is:

$$C_i(x', y', z) = \frac{Q_i(t)}{2\pi U \sigma_y \sigma_{z,i}} \exp\left(-\frac{y'^2}{2\sigma_y^2}\right) V_i(z, x') \cdot e^{-\lambda_i x' / U} \quad (20)$$

This produces qualitatively different spatial patterns by species:

- **Oak** (v_s small, λ small): Wide, far-reaching plumes. Concentration remains elevated hundreds of meters downwind. Dominates campus-wide background.
- **Pine** (v_s large, λ large): Narrow, rapidly decaying plumes. High concentration within 20–50 m of source, then sharp falloff. Creates localized “hot spots” near pine trees.
- **Redwood/Elm** (intermediate): Moderate plume extent, 50–150 m effective range.

Figure 2: Species-Dependent Plume Shapes
(Heavier pollen settles faster → higher concentration near source, shorter range)

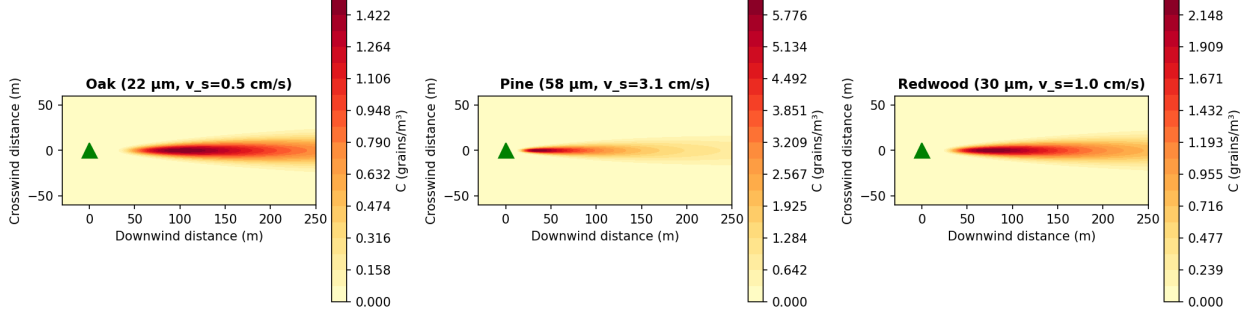


Figure 3: Species-dependent plume shapes due to different settling velocities. Oak pollen (small, light grains) produces wide, far-reaching plumes. Pine pollen (large, heavy grains) creates intense but localized concentration near the source tree.

2.7 Dispersion Coefficients

The crosswind (σ_y) and vertical (σ_z) standard deviations follow Pasquill-Gifford power-law relations [Gifford, 1961]:

$$\sigma_y = a_y(x')^{b_y}, \quad \sigma_z = a_z(x')^{b_z} \quad (21)$$

with coefficients (a, b) tabulated for atmospheric stability classes A–F (Table 2). Stability is estimated from wind speed and cloud cover using Turner’s method [Turner, 1964].

Table 2: Pasquill-Gifford dispersion coefficients.

Class	a_y	b_y	a_z	b_z
A	0.22	0.894	0.20	0.894
B	0.16	0.894	0.12	0.894
C	0.11	0.894	0.08	0.894
D	0.08	0.894	0.06	0.894
E	0.06	0.894	0.03	0.894
F	0.04	0.894	0.016	0.894

2.8 Building Wake Modifications

Campus structures create aerodynamic disturbances modifying local transport. We implement the Schulman-Scire downwash model [Schulman and Scire, 1980]:

2.8.1 Modified Dispersion Coefficients

For receptors influenced by a building of height H_b and width W_b :

$$\sigma_{y'} = \sqrt{\sigma_y^2 + \frac{W_b^2}{4\pi}} \quad (22)$$

$$\sigma_{z'} = \sqrt{\sigma_z^2 + \frac{H_b^2}{4\pi}} \quad (23)$$

2.8.2 Cavity Concentration

Within the recirculation zone ($x' \leq 3H_b$), turbulent mixing produces uniform concentration:

$$C_{cavity} = \frac{Q}{B_e \cdot H_b \cdot W_b \cdot U} \quad (24)$$

where $B_e \in [0.5, 0.7]$ is an empirical aerodynamic fraction.

2.8.3 Wake Damping

A binary mask $M_{wake}(x, y)$ identifies cells in building shadows, where wind velocity is damped by 60% and deposition enhanced by $2.5\times$.

2.9 Multi-Source Superposition

By linearity of Eq. (1), the total field from N sources is:

$$C_{total}(x, y) = \sum_{i=1}^N C_i(x, y) \quad (25)$$

evaluated via NumPy vectorized broadcasting over the full 120×120 spatial grid.

2.10 Building Footprint Masking

At $z = 1.5$ m within a building, there is no outdoor air exposure. Building rooftops detected from satellite imagery (Section 4.6) define a binary mask:

$$C_{masked}(x, y) = C_{total}(x, y) \cdot (1 - M_{building}(x, y)) \quad (26)$$

3 Phenological Emission Model

3.1 Temporal Gate Function

Species-specific seasonal emission is modeled as:

$$Q_i(t) = Q_{base,i} \cdot W_i \cdot \Gamma_i(t) \quad (27)$$

The three factors are:

- $Q_{base,i}$ (**Base Emission Rate**, grains s⁻¹): The maximum pollen production rate of species i at peak bloom under ideal conditions. This represents the intrinsic reproductive output of the plant and depends on species physiology, tree maturity, and canopy volume. Values are derived from aerobiological literature; for example, a mature Coast Live Oak may release ~500 grains/s at peak, while a Pine produces ~400 grains/s. Formally:

$$Q_{base,i} = \rho_i \cdot A_{canopy,i} \quad (28)$$

where ρ_i is the species-specific pollen flux density (grains s⁻¹ m⁻²) and $A_{canopy,i}$ is the canopy surface area. In our implementation, Q_{base} is assigned per-species from literature values (Table 3).

- W_i (**Allergen Potency Weight**, dimensionless, 0–5): A clinical severity index quantifying the allergenic impact of species i relative to other pollen types. This accounts for the fact that equal grain counts of different species produce vastly different immune responses. For instance, Oak pollen ($W = 4.0$ – 4.5) triggers severe rhinitis and asthma at lower concentrations than Pine pollen ($W = 2.0$), whose large grains are less immunologically reactive. The potency weight scales the effective emission to reflect health impact rather than raw grain count:

$$Q_{effective,i}(t) = Q_{base,i} \cdot W_i \cdot \Gamma_i(t) \quad (29)$$

Values are calibrated from clinical allergenicity databases and regional symptom prevalence data.

- $\Gamma_i(t)$ (**Temporal Gate Function**, dimensionless, 0–1): A smooth envelope controlling the seasonal on/off behavior of pollen release. Modeled as a truncated Gaussian:

$$\Gamma_i(t) = \begin{cases} \exp\left(-\frac{(t-t_{peak})^2}{2\sigma_t^2}\right) & t_{start} \leq t \leq t_{end} \\ 0 & \text{otherwise} \end{cases} \quad (30)$$

Here t is the Julian day of year (1–365), t_{start} and t_{end} define the pollination window boundaries, t_{peak} is the day of maximum bloom intensity ($\Gamma = 1$), and σ_t (days) controls the width of the blooming bell curve. Outside the window $[t_{start}, t_{end}]$, the tree is dormant and releases zero pollen regardless of weather.

Physical Interpretation. The product $Q_{base} \cdot W \cdot \Gamma(t)$ yields a time-varying “effective allergenic emission” in potency-weighted grains per second. On the peak bloom day ($\Gamma = 1$) of a high-potency species ($W = 4.5$), the effective source strength is $Q_{base} \times 4.5$. By mid-summer when $\Gamma = 0$ for all tree species, the effective emission is zero regardless of Q_{base} or W .

3.2 Species Phenological Parameters

Table 4 lists the temporal parameters for the six genera identified on campus, derived from Bay Area regional data [Bastl et al., 2016].

Table 3: Base emission rates and potency weights by species.

Species	Q_{base} (grains/s)	W (potency)
Valley Oak	550	4.5
Coast Live Oak	500	4.0
Coast Redwood	300	2.5
Pine	400	2.0
Chinese Elm	350	3.0
Western Sycamore	400	3.5

Table 4: Species phenological parameters.

Species	t_s	t_e	t_p	σ_t	W
Valley Oak	60	120	95	21	4.5
Coast Live Oak	60	120	91	21	4.0
Redwood	32	91	60	21	2.5
Pine	91	181	135	30	2.0
Chinese Elm	244	305	274	21	3.0
Sycamore	74	121	100	18	3.5

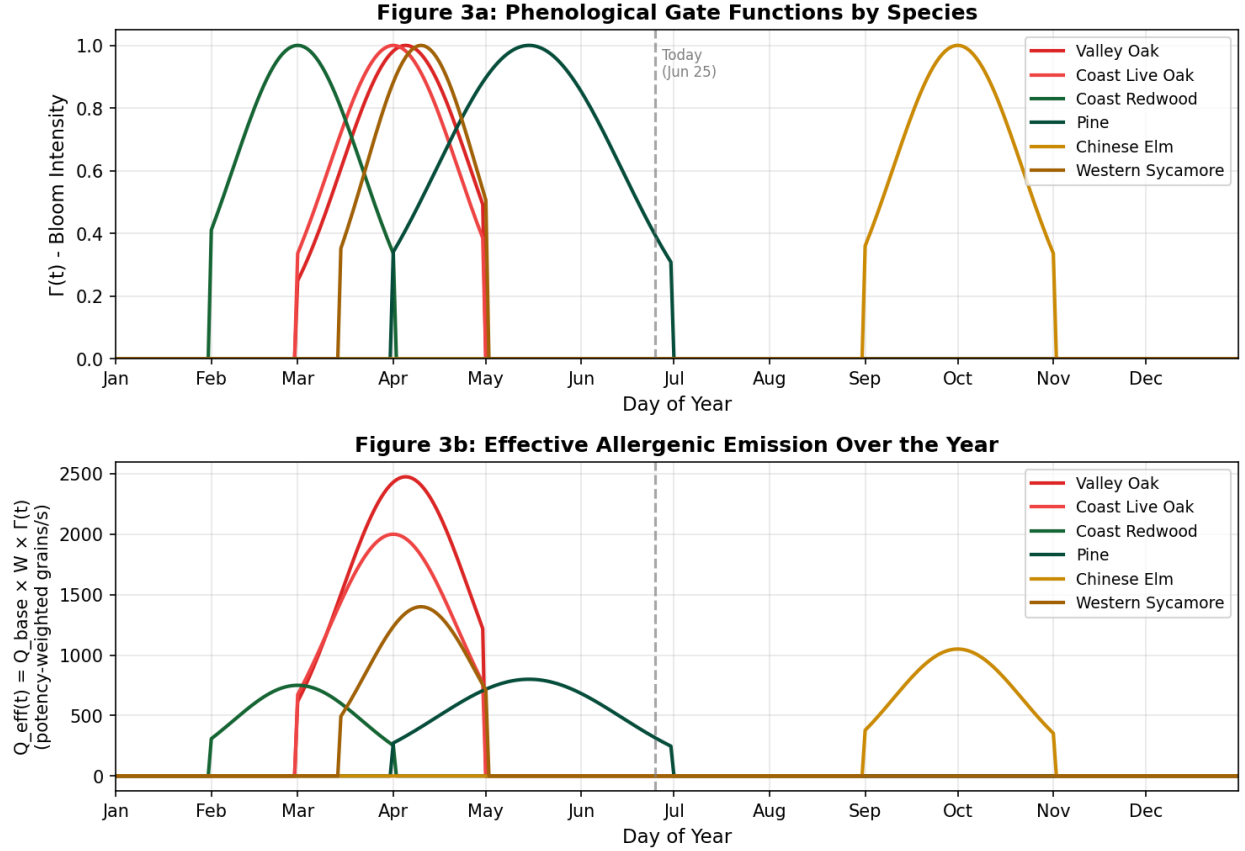


Figure 4: Phenological emission model. (a) Temporal gate functions $\Gamma_i(t)$ showing blooming intensity across the year for each species. (b) Effective allergenic emission $Q_{eff}(t) = Q_{base} \times W \times \Gamma(t)$. The dashed line marks today (June 25), when most tree species are dormant and only Pine retains weak late-season emission.

4 Satellite-Based Vegetation Detection

4.1 Image Acquisition

High-resolution satellite imagery (0.5 m/pixel) is obtained from the Esri World Imagery REST export API, covering the full campus (557 m $\times$ 637 m) at 1600 $\times$ 1398 pixels.

4.2 Calibrated Color Segmentation

Tree canopy pixels are classified using thresholds calibrated from 771 human-verified positive samples:

$$\text{is_tree} = (g_e > 4) \wedge (35 < \bar{I} < 155) \wedge (r < 165) \wedge (b < 125) \quad (31)$$

where $g_e = g - (r + b)/2$ is the green excess and $\bar{I} = (r + g + b)/3$ the mean brightness.

4.3 Watershed Segmentation

Adjacent canopies are separated using marker-controlled watershed [Beucher and Meyer, 1992]:

1. Binary mask is eroded (3 iterations) to generate seed markers
2. Euclidean distance transform computed on the mask
3. Watershed algorithm segments using inverted distance as elevation

4.4 Peak-Based Detection

An alternative pipeline computes a “tree-ness” score:

$$S(x, y) = \max(g_e, 0) \cdot \frac{\max(155 - \bar{I}, 0)}{120} \quad (32)$$

convolved with a Gaussian ($\sigma = 4$ px). Local maxima with minimum spacing of 7 pixels identify individual tree centers. Canopy radius is estimated from the score falloff profile.

4.5 Human-in-the-Loop Calibration

The detection undergoes iterative refinement:

1. Automated detection identifies candidates
2. Users delete false positives via interactive map
3. Verified positions persist and inform threshold updates
4. Three rounds achieved $< 5\%$ false positive rate

4.6 Building Rooftop Detection

Building footprints are identified as connected regions satisfying:

$$(\text{saturation} < 30) \wedge (\bar{I} > 160) \wedge (g_e < 8) \quad (33)$$

with minimum area 800 pixels (≈ 20 m²).

5 Exposure Assessment

5.1 Path-Integrated Dose

Cumulative exposure along walking path γ at speed $v_w = 1.4$ m/s:

$$D = \int_{\gamma} C_{total}(\mathbf{x}(s)) \cdot \frac{ds}{v_w} \approx \sum_k C(\mathbf{x}_k) \cdot \Delta t_k \quad (34)$$

5.2 Clinical Risk Mapping

Dose is mapped to actionable risk levels: Low ($D < 50$), Moderate (50–200), High (200–500), Very High (> 500 grain·s/m³).

6 Results

6.1 Detection Performance

After three calibration rounds, the system identifies 1,018 trees: 414 Coast Live Oak, 239 Redwood, 159 Valley Oak, 110 Pine, 76 Sycamore, and 20 Chinese Elm.

6.2 Seasonal Concentration

Using a fixed scale (500 grains/m³ = spring peak):

- **April 1** (day 91): $C_{max} = 273$ grains/m³ (55% of scale); 8 active species
- **June 25** (day 176): $C_{max} = 30$ grains/m³ (6%); Pine only
- **January** (day 15): $C_{max} \approx 0$; all species dormant

6.3 Spatial Patterns

The concentration field exhibits: (i) plume elongation along prevailing wind; (ii) zero concentration over building footprints; (iii) enhanced accumulation in building wake zones; (iv) elevated concentrations in oak-dense corridors.

7 Discussion

The Gaussian plume approach provides a computationally efficient (< 500 ms per field computation) approximation suitable for real-time interactive applications. Key limitations include the assumption of spatially uniform wind (neglecting building-channeled flows that would require CFD), the RGB-only species classification (improved with hyperspectral data), and absence of ground-truth pollen measurements for quantitative validation.

The human-in-the-loop calibration workflow proved essential: automated detection alone achieved approximately 75% precision, improved to $> 95\%$ after three interactive correction rounds. This suggests that for campus-scale applications, semi-automated approaches combining algorithmic detection with targeted human verification are more practical than fully autonomous methods.

The fixed annual scale reveals that allergy risk communication should emphasize seasonal context: a June reading of 30 grains/m³ might alarm students if presented without reference to the 500 grains/m³ spring peak that constitutes the actual high-risk period.

8 Conclusion

We demonstrated a complete pipeline for hyperlocal pollen dispersion modeling, from satellite-based source identification to physics-based atmospheric transport to actionable exposure assessments. The system resolves concentration gradients at 5-meter scale across a 25-hectare campus with 1,018 individually mapped pollen sources, enabling allergy-sensitive students to make informed daily routing decisions based on current meteorological conditions and seasonal phenology.

Data Availability

Source code and campus data are available at the project repository. The tree inventory is derived from the PAUSD 2009 arborist survey supplemented by satellite detection.

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